MILITARY SPECIFICATION

CARTRIDGE, 20MM, DUMMY, DUMBELL, GUN DRIVE DEACTIVATOR M51A1E2

1. SCOPE

1.1 This specification covers the Cartridge 20MM, Dummy, Dumbbell, Gun Drive Deactivator M51A1E2 for use in Guns, Automatic, 20MM, M61 and M39.

2. APPLICABLE DOCUMENTS

3.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–109 —Quality Assurance Terms and Definitions

DRAWINGS

ORDNANCE CORPS

D7259085 —Cartridge, 20MM, dummy, dumbbell, gun drive deactivator, M51-A1E2

F7258942 —Method of Packing Cartridges (200), M50 Series, 20MM, in Bulk, in Box, Ammunition, T46

D7259079 —Marking on Box, Ammunition, T46, for Cartridges, 20MM, Dummy

PUBLICATIONS

ORD–M608–11 —Procedures and Tables for Continuous Sampling by Attributes

ORD–M608–14 —Ammunition Ballistic Acceptance Test Methods, Part VIII

IEL–7259085 —Index of Inspection Equipment Lists for: Cartridge 20MM, Dummy, Dumbbell, Gun Drive Deactivator, M51A1E2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein.

ASTM–A–219–58 —Standard Methods of Test for Local Thickness of Electrodeposited

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with applicable drawings and specifications.

3.2 **Parts.** The parts and the assembled cartridge shall comply with requirements of applicable drawing and specifications.

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3.3 **Projectile extraction.** The cartridge assembly shall withstand a 3,900 pound tension force without separation of the projectile from the cartridge case when tested as specified in 4.5.1.

3.4 **Preproduction samples.** Preproduction samples shall be required prior to start of regular production (see 4.3.)

3.5 **Workmanship.** The cartridge assembly shall be clean, completely finished, and free from all surface defects such as burrs, fins, slivers, rough edges, chatter marks, scores, dents, and scratches. The cleaning methods used shall not be injurious to the cartridge assembly nor shall the cartridge assembly be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL–A–2550 apply to this specification. Reference will be made to MIL–STD–109 for definition of inspection terms used herein.

4.3 **Preproduction sample.** Prior to the start of regular production, the contractor shall submit a preproduction sample manufactured by the same methods and equipment that will be utilized in performing the work under the contract. The contracting officer will furnish instructions for shipment of the sample to a designated Government laboratory for examination and test.

4.3.1 The preproduction sample shall consist of twenty-five (25) cartridge assemblies. The sample will be sent to and examined by a Government laboratory for compliance with the requirements of the applicable drawings and this specification.

4.3.2 Results of the examination and test will be furnished to the contractor, Approval or disapproval of the preproduction sample will be given by the responsible procuring activity. Instructions for the submission of new sample(s) in case of disapproval of the original sample will be issued by the procuring activity. Regular production shall not proceed until the preproduction sample has been approved.

4.4 **Sampling.**

4.4.1 *Lot formation.* A lot shall consist of cartridge assemblies produced in accordance with the same drawings and drawing revisions, and same specification and specification revision.

4.4.2 *Examination.*

4.4.2.1 Lot sampling and acceptance criteria will be in accordance with MIL–STD–105 or ORD–M608–11, as appropriate. Judgment for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.2):

4.4.2.1.1 *Individual defect characteristic basis.*

Major defects	0.25%
Minor defects	0.40%

4.4.2.1.2 *Defect characteristic class basis.*

Major defects	1.50%
Minor defects	2.50%

Individual defect characteristic AQL's associated with the class basis will be

Major defects	0.40%
Minor defects	0.65%

4.4.3 *Test Inspection.* The tests specified in table I will be performed in accordance with the methods specified in 4.5. Acceptance criteria are as specified below:

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TABLE I

Test	Part
Projectile extraction	Cartridge Assembly
Coating thickness	Cartridge Assembly

4.4.3.1 *Projectile extraction.* A double sampling plan using the method of MIL–STD–105 will be used in this test. Sample sizes will be those associated with applicable sampling tables and procedures. Acceptance criteria will be those given for the applicable sampling scheme using an AQL of 0.65%. If a cartridge case projectile separation occurs at less than the prescribed tension force the cartridge will be classed defective. However, if a separation occurs at less than the prescribed tension force by rupturing of the cartridge case, no penalty to the lot will be assigned.

4.4.3.2 *Plating thickness.* A double sampling plan using the method of MIL–STD–105 will be used in this test. Sample sizes will be those associated with applicable sampling tables and procedures. Acceptance criteria will be those given for the applicable sampling scheme using an AQL of 0.65%. Any cartridge assembly which does not have a minimum protective coating will be classed defective.

4.4.4 *Classification of defects.*

4.4.4.1 *Cartridge assembly* (see Dwg. D72-59085).

Categories and defects	*Method of inspection*
Critical	
None defined.	
Major	
101. Case plug missing	Visual
102. Profile and alignment, max.	Gage
103. Case-projectile crimp missing or not full 360°	Visual
104. Weight	Weight
105. Total length	Gage
106. Case plug stake(s) missing	Visual
107. Depth of case plug, min.	Gage
108. Length, case head to centerline of crimp	Gage
109. Length, case head to projectile body relief	Gage
110. One or more spot welds missing	Visual
Minor	
201. Depth of case plug, max	Gage
202. Plating missing or damaged	Visual*
203. Grease, oil or other foreign matter	Visual
204. Scratches, dents, buckles, folds	Visual
205. Marking incomplete or unidentifiable	Visual
206. Diameter of bourrelet	Visual

* See 4.4.5.1.

4.4.4.2 *Bulk pack* (see Drawing F7258942).

Categories and defects	*Method of inspection*
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Incorrect number of cartridges in container	Visual
202. Missing or improper fillers or tubes	Visual
203. Improper packing of cartridges	Visual

4.4.4.3 *Closed box* (see Dwg. F7259079).

Categories and defects	*Method of inspection*
Critical	
None defined.	
Major	
101. Incorrectly closed and secured box	Visual

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Categories and defects	Method of inspection
102. Missing or unserviceable handle	Visual
103. Missing or incorrect or unidentifiable ammunition lot number or FSN	Visual
104. Foreign matter	Visual
Minor	
205. Marking, other than ammunition lot number or FSN, missing or incomplete	Visual
206. Marking touched up or repaired with non-permissible material (crayon, chalk, etc.)	Visual

4.4.5 *Defect standards.*

4.4.5.1 Any cartridge assembly which is not plated or which shows blistered, scaly or spotty plating will be classed defective.

4.4.6 *Inspection equipment.*

4.4.6.1 Government verification inspection will be accomplished with the use of gages and equipment specified in the Index of Inspection Equipment Lists for: Cartridge, 20mm, dummy, dumbbell, gun drive deactivator, M51A1E2, IEL–7259085.

4.5 Test methods.

4.5.1 *Projectile extraction.* The cartridge of the test sample will be tested as specified in Section II of ORD–M608–14, Ammunition Ballistic Acceptance Test Methods, Part VIII. The tension force will be applied at the specified rate until the indicated load of 3,900 pounds is reached. The cartridges of the sample will be held at this load for ten (10) seconds after which the load will be released and the cartridge examined. Cartridge assemblies which are not damaged or distorted and which are free of the defect characteristics listed in 4.4.4.1 may be returned to the lot from which they were selected.

4.5.2 *Plating thickness.* Method of test will be as specified in ASTM Standard, Part 3, A–219–58 and Appendix I thereto.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A and B.* Cartridges shall be packed in accordance with Drawing F7258-942.

5.2 Marking. Marking on ammunition packs shall be as specified on Drawing D7259079.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this purchase description.

(b) Place of inspection, if not at place of manufacture.

(c) Packing and marking required.

6.2 AQL's. The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is authorized to minimize inspection agency administrative burden which might result, from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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